A PROJECT REPORT

ON

“SafeSpace”

SUBMITTED IN PARTIAL FULFILMENT OF THE REQUIREMENTS FOR THE DEGREE OF

(BACHELOR OF COMPUTER SCIENCE)

SEM V

BY

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UNDER THE GUIDANCE OF

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SONOPANT DANDEKAR ARTS, V.S. APTE COMMERCE AND M.H. MEHTA SCIENCE COLLEGE, PALGHAR

UNIVERSITY OF MUMBAI

2026–2027

Sonopant Dandekar Shikshan Mandali's

Sonopant Dandekar Arts, V.S. Apte Commerce and M.H. Mehta Science College, Palghar.

Department of Computer Science

CERTIFICATE

This is to certify that **Ms. Prajakta Anil Mane**, Seat No. **510512514**, a student of T.Y.B.S.C(Computer Science), Semester V, has successfully completed the project titled **"SafeSpace"** under the guidance of Ms. Bhakti Patil in partial fulfilment of the requirements for the award of the degree of Bachelor of Science in Computer Science under the University of Mumbai during the academic year 2026 – 2027.

This project is the original work of the student and has been carried out under my guidance and supervision. It has not been submitted to any other university or institution for the award of any degree or diploma.

Project Guide	**Head of Department**	**External Examiner**

Date:

Place: College Seal

DECLARATION

I, **Prajakta Anil Mane**, hereby declare that the project entitled "**SafeSpace"** submitted in the partial fulfilment of the requirements for the award of **Bachelor of Science in Computer Science** during the academic year 2026-2027, is my original work and the project has not formed the basis for the award of any degree, or any other similar title.

Signature of the student:

Place:

Date:

ACKNOWLEDGEMENT

I would like to express my sincere gratitude to everyone who has supported and guided me throughout the completion of my project titled **“SafeSpace”.**

First and foremost, I would like to thank my project guide, **Prof. Bhakti Raut**, for her valuable guidance, continuous encouragement, and support throughout the development of this project. Her suggestions and guidance helped me understand the different stages of project development and complete the work successfully.

I am also thankful to the **Department of Computer Science and Sonopant Dandekar Arts, V.S. Apte Commerce and M.H. Mehta Science College, Palghar**, for providing me with the opportunity and necessary support to undertake this project as part of my Bachelor of Science in Computer Science.

I would like to express my gratitude to all the teachers and faculty members who provided their valuable suggestions and encouragement during the course of this project.

I am grateful to my family and friends for their constant motivation, patience, and support throughout the completion of this work.

Finally, I would like to thank everyone who directly or indirectly contributed to the successful completion of my project. This project has been a valuable learning experience that helped me improve my technical knowledge and understand how technology can be used to address important social issues.

Place: Palghar

Date:

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ABSTRACT

SafeSpace is a web-based project designed to provide discreet access to support and safety-related information for individuals experiencing domestic abuse. Many victims face difficulties seeking help because their activities may be monitored by abusers. Visiting a website that openly identifies itself as a domestic violence support platform may expose their intention and create additional risks. SafeSpace aims to address this problem by providing a discreet and user-friendly digital platform.

The main feature of SafeSpace is its neutral-looking interface, which initially appears as a normal recipe website. This design helps avoid immediately revealing the actual purpose of the platform. Through its support interface, users can access emergency help information, safety planning resources, legal information, secure contact options, and anonymous chat functionality. The project also includes a Quick Exit feature intended to help users leave the support interface quickly when required.

The website is developed using HTML, CSS, and JavaScript. HTML provides the structure of the website, CSS manages its appearance and responsive layout, and JavaScript enables interactive features and user navigation. The project is deployed using Netlify, making it accessible through a web browser. The current version follows a frontend-oriented architecture without a dedicated custom backend or database.

The primary objective of SafeSpace is to combine web technology, discreet interface design, and easy access to support resources to address an important social problem. The project demonstrates how a carefully designed website can help improve access to safety information while emphasizing privacy and usability. Future improvements may include integration with NGOs, emergency services, AI-assisted support, additional language options, and advanced safety-planning features.

Keywords: SafeSpace, Domestic Abuse, Web Application, Privacy, Emergency Help, Safety Planning, HTML, CSS, JavaScript.

Chapter 1. Introduction

1. Introduction

Domestic abuse continues to be one of the most alarming social issues in today's world, and despite countless awareness programs and legal reforms, a large number of victims still suffer in silence. The situation is made worse because abusers often maintain strict control over their victims, keeping a close watch on their daily activities, monitoring their phone usage, and even tracking their internet history. In such a hostile environment, the simple act of searching for help online or dialling a helpline number can put the victim in greater danger. Although there are many helpline services, mobile applications, and NGO support portals already available, these platforms are usually easy to identify, leave digital traces, and lack the essential element of discretion that victims desperately need.

Traditional help websites and support applications may clearly identify themselves as domestic-violence or emergency-support platforms. In a situation where a victim's device or internet activity is being monitored, openly accessing such a website can create additional risk.

SafeSpace is designed as a discreet web-based support platform. Its primary concept is to make the initial website appear like a normal and harmless recipe website. The actual website currently displays **"Fresh Recipes"**, including recipes, cooking tips and an About section, rather than immediately presenting itself as a support website. safespace-web.netlify.app

Behind this neutral interface, the project provides access to support-oriented features such as emergency help, resources, safety planning, legal information, secure contact and anonymous chat. safespace-web.netlify.app

From a technological perspective, the project is intentionally developed using only frontend technologies — HTML to structure the content, CSS to style and disguise the interface, and JavaScript to bring in interactivity and secret functionalities such as quick exit and code triggers. The deployment on Netlify ensures that the platform is lightweight, secure, and easily accessible without requiring complex backend systems that may compromise privacy or store sensitive user information. By adopting this approach, SafeSpace ensures that it leaves no digital footprint, no browsing history, and no trace of interaction that could alert an abuser.

Chapter 2. Problem Statement

2.1 Social Problem

Domestic abuse victims may live under continuous monitoring and control. In such circumstances, openly searching for domestic-abuse support or visiting an identifiable support website may attract unwanted attention.

Existing digital support systems can therefore face a usability and safety challenge: the victim may need help, but the appearance of the support platform itself can reveal their intention.

The project documentation identifies the lack of discretion in conventional support platforms as a major problem. It explains that visible help websites, registration requirements, notifications and digital traces may create additional risks for victims living under surveillance.

2.2 Institutional Problem

Institutions, NGOs and support organizations require digital channels through which people can access information and assistance. However, a platform that visibly identifies itself as a domestic-abuse support system may not be suitable for every user situation.

2.3 Proposed Solution

SafeSpace addresses this problem by providing:

- A neutral-looking initial interface
- Discreet access to support features
- Emergency-help information
- Safety-planning information
- Legal information
- Secure contact functionality
- Anonymous chat
- Quick exit functionality
- Privacy-oriented design

The project specifically defines the problem as creating a digital platform through which domestic-abuse victims can access support in a safe, disguised and anonymous manner.

Chapter 3. Objectives

3.Objectives

The major objectives of SafeSpace are:

1. To provide a discreet digital environment for people seeking support.
2. To avoid immediately revealing the purpose of the website.
3. To provide emergency-support information.
4. To provide safety-planning resources.
5. To provide legal-rights information.
6. To provide discreet communication facilities.
7. To provide a quick-exit mechanism.
8. To maintain a lightweight frontend architecture.
9. To provide responsive access across different devices.
10. To create an easy-to-use and privacy-oriented interface.

The project's original objectives specifically include disguised entry, quick exit, silent communication, coded messages and access to legal rights and safety information.

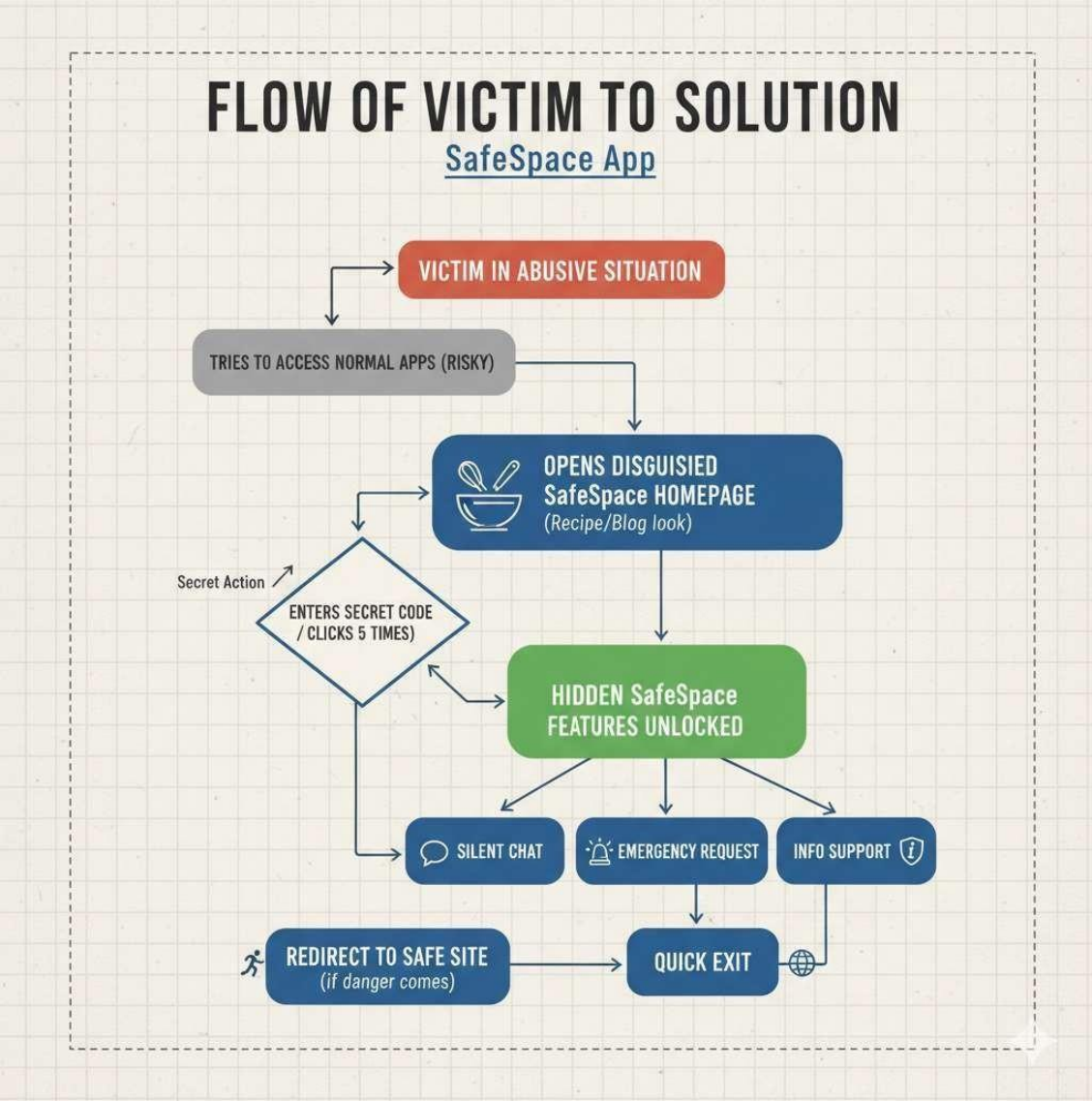
3.1 Scope

The scope of SafeSpace includes:

- Disguised website interface
- Emergency help
- Safety planning
- Legal information
- Support resources
- Secure contact
- Anonymous chat
- Responsive web interface
- Privacy-oriented design
- Future integration with NGOs and emergency services

The project documentation also identifies future possibilities such as NGO integration, emergency responders, AI-driven support and geolocation-based SOS functionality.

How it works:


FLOW OF VICTIM TO SOLUTION
SafeSpace App
VICTIM IN ABUSIVE SITUATION
TRIES TO ACCESS NORMAL APPS (RISKY)
OPENS DISGUISIED
SafeSpace HOMEPAGE
(Recipe/Blog look)
Secret Action
ENTERS SECRET CODE
/ CLICKS 5 TIMES)
HIDDEN SafeSpace
FEATURES UNLOCKED
SILENT CHAT
EMERGENCY REQUEST
INFO SUPPORT
REDIRECT TO SAFE SITE
(if danger comes)
QUICK EXIT

Chapter 4. Feasibility Study

4.Feasibility Study

A feasibility study determines whether the SafeSpace project can realistically be developed and operated.

4.1 Technical Feasibility

SafeSpace is technically feasible because it can be implemented using commonly available web technologies.

Technologies

- **HTML** – structure of web pages
- **CSS** – styling and responsive design
- **JavaScript** – interaction and hidden functionality
- **Web Browser** – client-side execution
- **Netlify** – deployment

The project documentation explicitly states that SafeSpace uses HTML, CSS and JavaScript and is deployed using Netlify.

Technical Advantages

- Lightweight
- Easy to maintain
- No complex server infrastructure for the current design
- Accessible through standard web browsers
- Responsive
- Easy deployment

4.2 Economic Feasibility

The project is economically feasible because its current architecture primarily uses frontend technologies.

Cost Factors

Component	Expected Result
HTML	Free
CSS	Free
JavaScript	Free

Development Tools	Free / Low Cost
Web Browser	Existing Software
Netlify Development	Depends on selected hosting plan
Database	Not required in current architecture

The project avoids a complex backend/database architecture in its current design, reducing infrastructure requirements.

4.3 Operational Feasibility

SafeSpace is operationally feasible because users can access it through a standard web browser.

The website's visible interface includes navigation such as Home, Recipes, Cooking Tips and About, creating a normal recipe-site appearance. safespace-web.netlify.app

The support area provides options including:

- Immediate Help
- Resources & Information
- Safety Planning
- Legal Information
- Secure Contact
- Anonymous Chat safespace-web.netlify.app

Chapter 5. Requirement Engineering

5. Requirement Engineering

Requirement engineering identifies what the SafeSpace system must provide.

5.1 Functional Requirements

FR1 – Disguised Homepage

The system shall initially display a normal-looking recipe website.

FR2 – Support Access

The system shall provide access to the SafeSpace support section.

FR3 – Emergency Help

The system shall display emergency-support information.

The current website contains an Immediate Help section and a listed National Domestic Violence Hotline. safespace-web.netlify.app

FR4 – Safety Planning

The system shall provide safety-planning functionality.

FR5 – Legal Information

The system shall provide information about legal rights and available options.

FR6 – Secure Contact

The system shall provide a mechanism for contacting support organizations.

FR7 – Anonymous Chat

The system shall provide an anonymous communication interface.

FR8 – Quick Exit

The system shall allow the user to quickly leave the support interface.

FR9 – Resource Information

The system shall provide safety and support resources.

5.2 Non-Functional Requirements

Privacy

The system should minimize exposure of sensitive user information.

Usability

The interface should be simple enough to understand and operate quickly.

Responsiveness

The website should work across desktops, tablets and smartphones.

Performance

Pages should load quickly because the application is lightweight.

Maintainability

HTML, CSS and JavaScript components should be easy to modify.

Availability

The website should be accessible through the internet when deployed.

Security-Oriented Design

The interface should avoid unnecessarily exposing the user's intention to seek support.

Compatibility

The website should work with modern web browsers.

The original project documentation specifically identifies responsive multi-device access and frontend-only architecture as part of the project scope.

Chapter 6. Software Development Life Cycle Planning

6. Software Development Life Cycle Planning

SafeSpace can be developed using an **iterative SDLC approach**.

SDLC Phases

Phase 1 – Requirement Analysis

Identify the problems faced by domestic-abuse victims and define the required features.

Phase 2 – System Design

Design the website structure, navigation, disguised interface and support modules.

Phase 3 – Development

Implement the website using HTML, CSS and JavaScript.

Phase 4 – Testing

Test navigation, responsiveness, hidden functionality, emergency features and user interface.

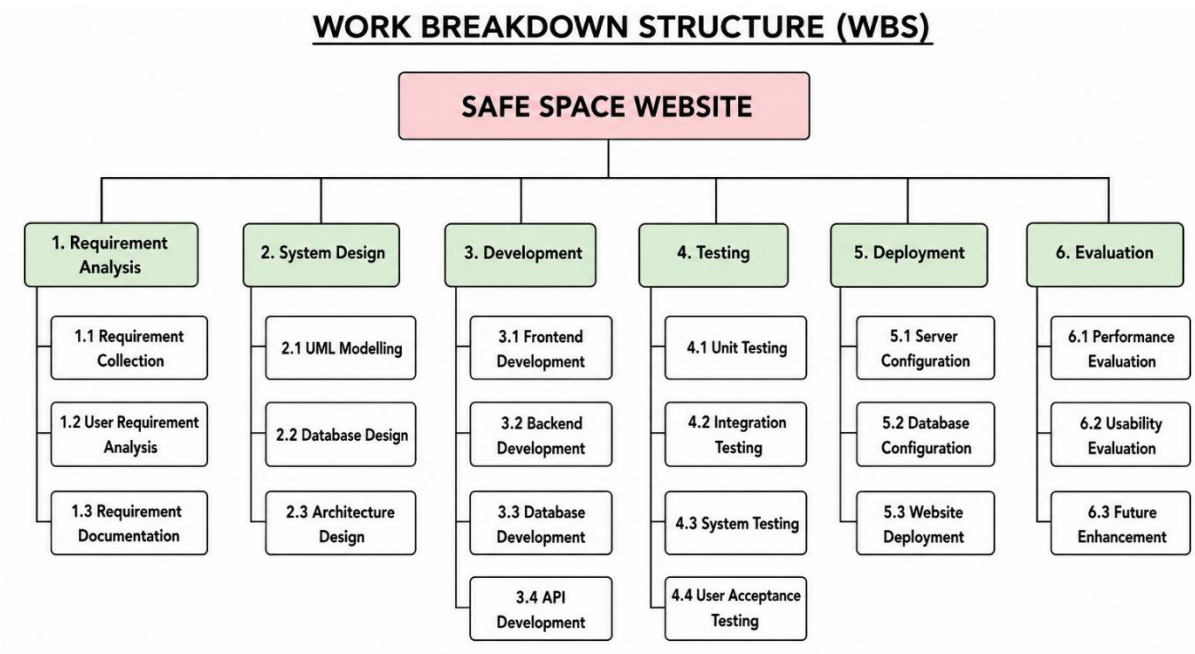
Phase 5 – Deployment

Deploy the completed website using Netlify.

Phase 6 – Evaluation

Evaluate usability, functionality, responsiveness and privacy-oriented design.

6.1 Work Breakdown Structure (WBS)



6.1 WORK BREAKDOWN STRUCTURE(WBS)

6.2 Project Timeline

Gantt Chart

	Week 1	Week 2	Week 3	Week 4	Week 5	Week 6
Requirement Analysis						
System Design						
UML Design						
Frontend Development						
Backend Development						
Database Development						
Testing						
Deployment						
Evaluation						
Documentation						

6.2 Project/Gantt Chart

Chapter 7. System Model Using UML

7.1 Use Case Diagram

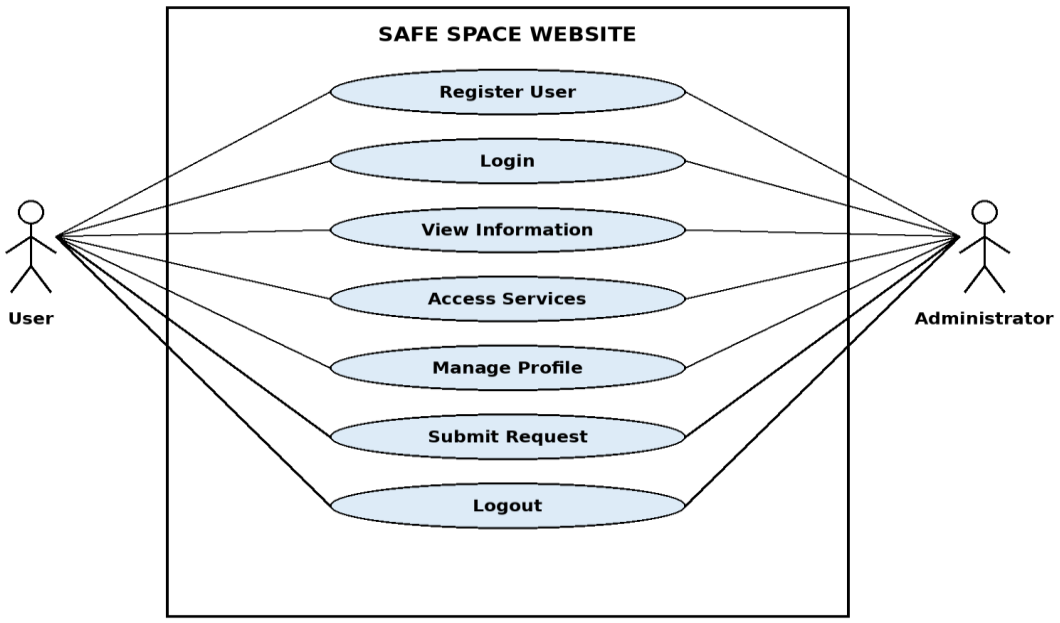
The Use Case Diagram represents the interaction between users and the SafeSpace system. It shows the different functions available to users through the website. Users can access emergency help, support resources, safety planning, and legal information.

User / Victim

Use Cases

- Open SafeSpace
- View Neutral Homepage
- Access Support Area
- View Emergency Help
- View Resources
- Create Safety Plan
- View Legal Information
- Contact Support
- Use Anonymous Chat
- Quick Exit

Diagram Structure



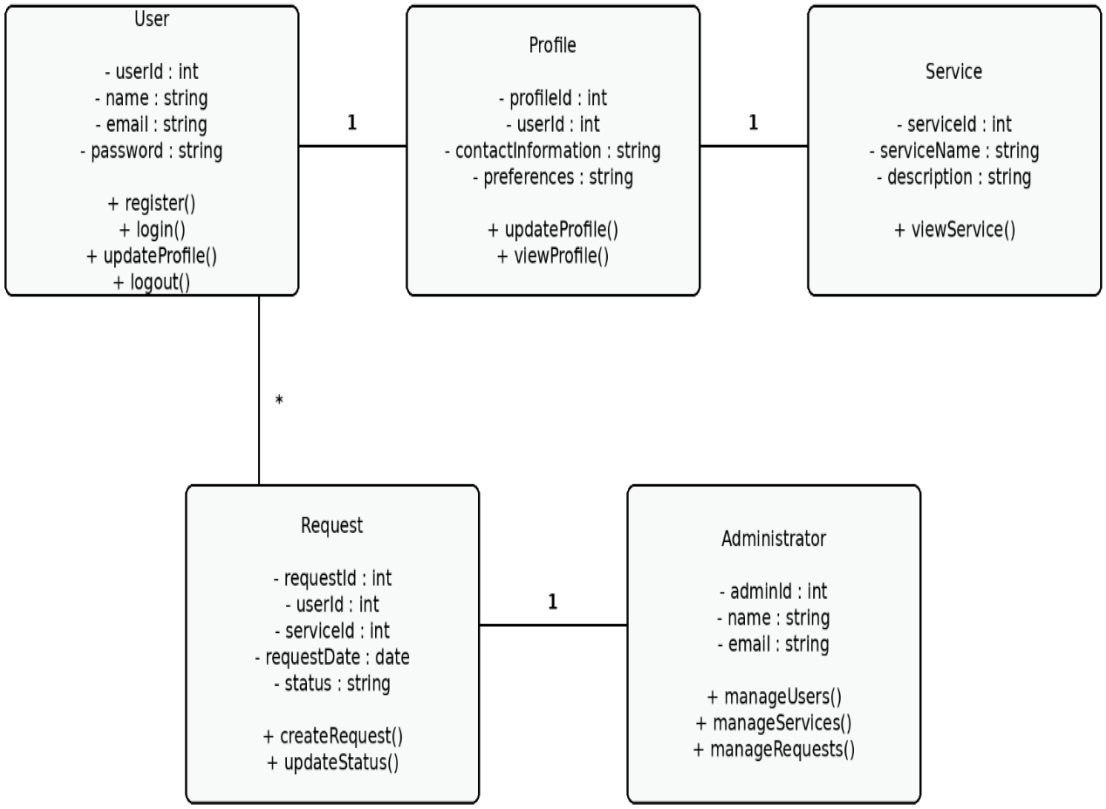
7.1 Use Case Diagram

The actual website exposes emergency help, resources, safety planning, legal information,secure contact and anonymous chat options. safespace-web.netlify.app

7.2 Class Diagram

The Class Diagram represents the main classes of the SafeSpace system and describes their attributes and operations. It shows the relationships between different classes used to represent the website's structure. These classes may represent features such as support resources, safety planning, and legal information. The diagram helps explain the structural organization of the system and the relationships between its elements.

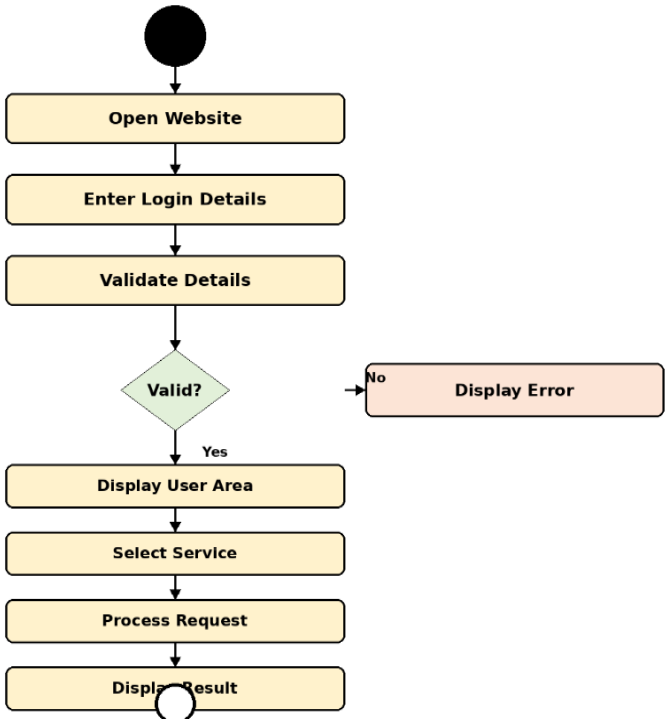
Possible logical classes for the current system:



7.2 Class Diagram

7.3 Activity Diagram

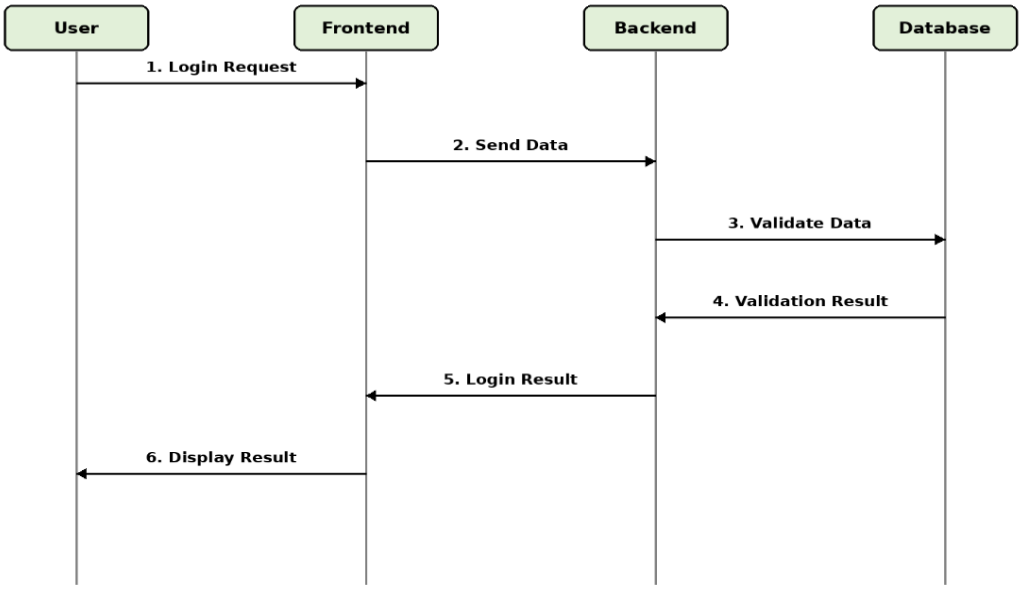
The Activity Diagram represents the workflow of the SafeSpace system. It illustrates the sequence of activities performed when a user accesses and navigates through the website. The process begins when the user opens the website and selects a required feature. The system then provides access to the relevant information or resources. This diagram explains the flow of activities from the beginning to the end of a process.



7.3 Activity Diagram

7.4 Sequence Diagram

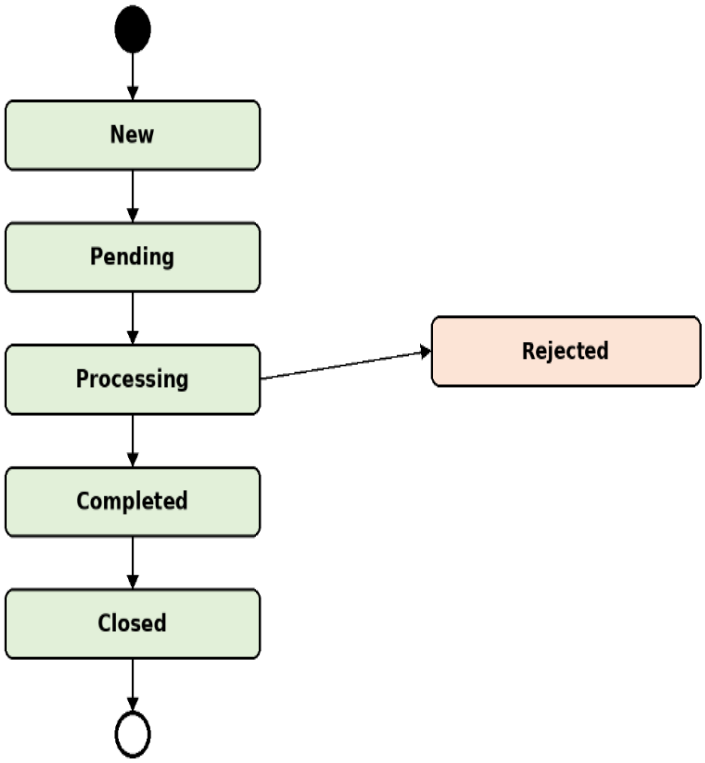
The Sequence Diagram represents the interactions between the user and the SafeSpace system in chronological order. It shows how actions and responses occur during website usage. The user accesses the website and selects a particular feature, after which the system responds by displaying the relevant interface or information. The diagram illustrates the order of interactions and helps explain how the user accesses the available website features.



7.4 Sequence Diagram

7.5 State Diagram

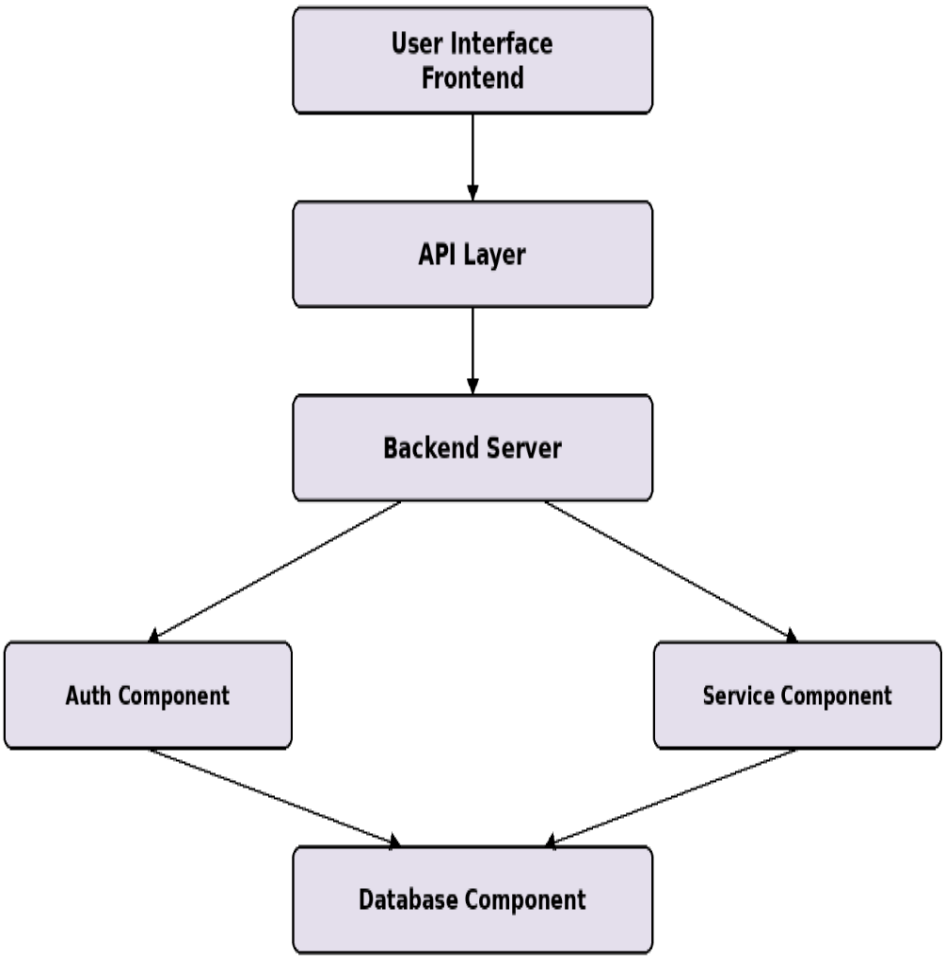
The State Diagram represents the different states of the SafeSpace system and the transitions between them. It illustrates how the website's state changes in response to user actions. For example, a user may begin at the homepage and then navigate to a support section or another available feature. The diagram helps explain the possible states of the system and how users move between different sections of the website.



7.5 State Diagram

7.6 Component Diagram

The Component Diagram represents the main software components of the SafeSpace system and their organization. The website uses HTML to structure content, CSS to manage presentation, and JavaScript to support interactive functionality. These technologies work together to provide website navigation and access to available features. The diagram gives an overview of the software components and helps explain the organization of the SafeSpace website.

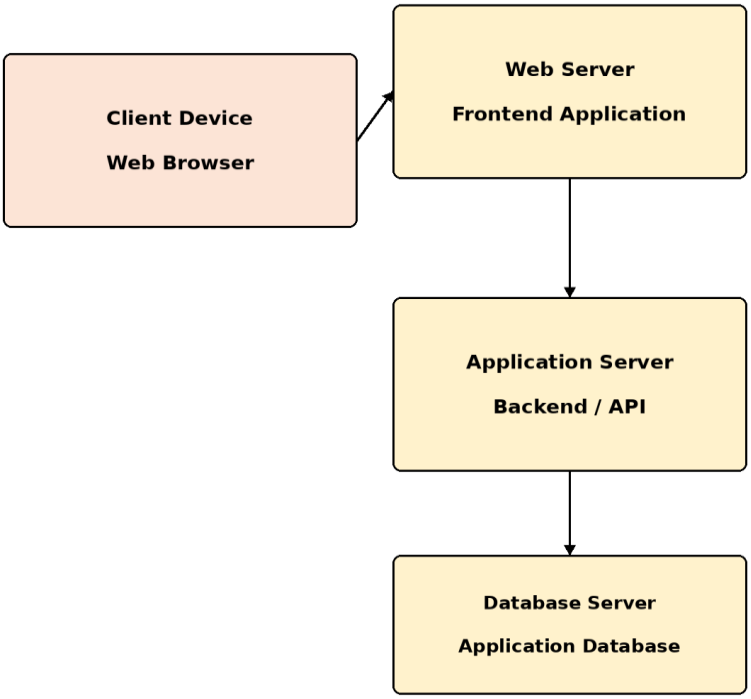


7.6 Component Diagram

The project source describes the architecture as frontend-only using HTML, CSS and JavaScript.

7.7 Deployment Diagram

The Deployment Diagram represents the deployment environment of the SafeSpace website. Users access the website through a web browser using an internet connection. The website is developed using HTML, CSS, and JavaScript and is deployed on Netlify. This diagram illustrates the environment in which the website operates and explains how users can access it online.

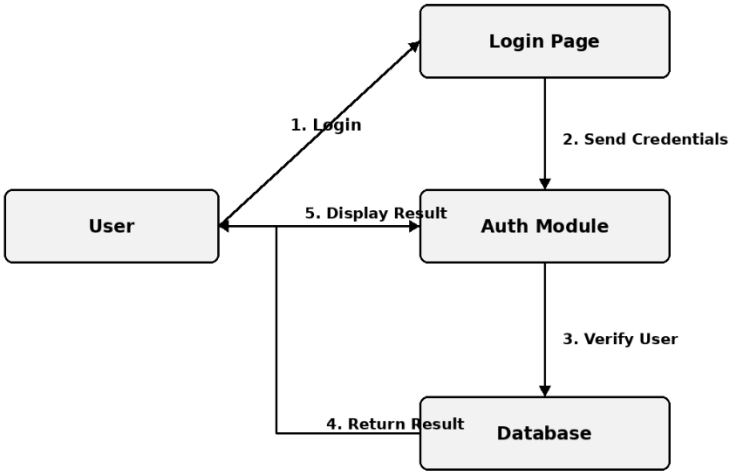


7.7 Deployment Diagram

The supplied project documentation identifies Netlify as the deployment platform.

7.8 Communication Diagram

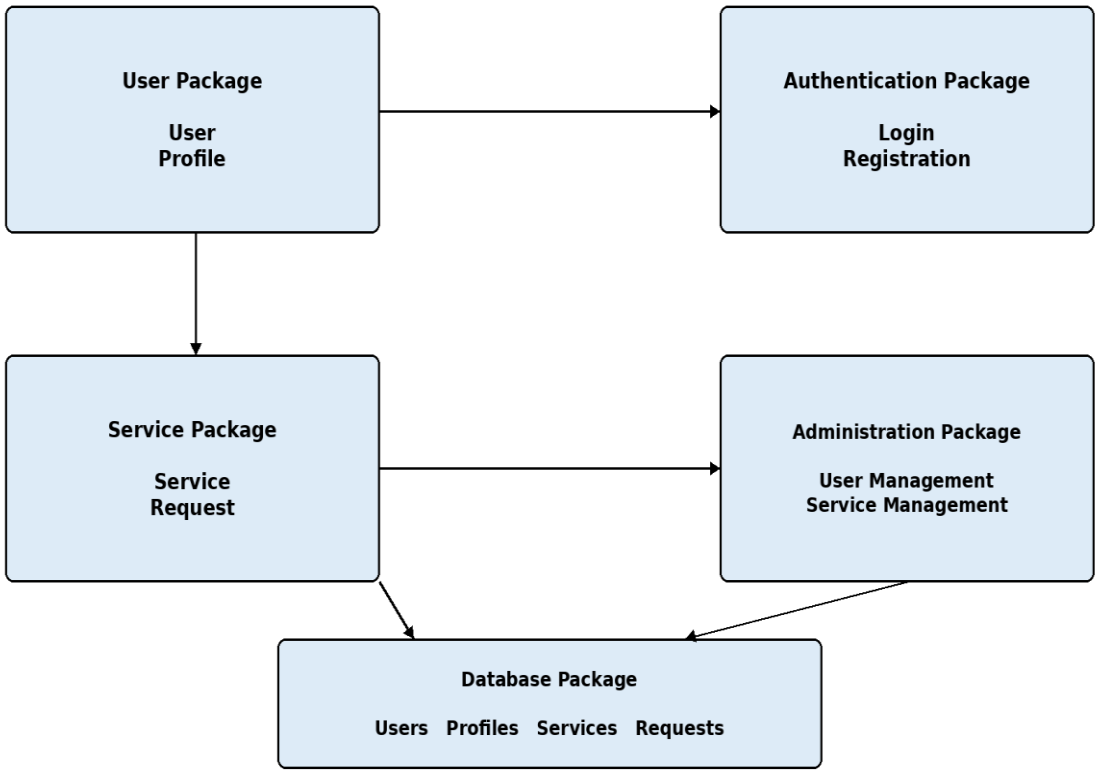
The Communication Diagram represents the communication and relationships between the interacting elements of the SafeSpace system. It illustrates how user actions are connected with the relevant website functionality. When users select a feature, they can access the corresponding support information or resources available on the website. The diagram provides an overview of the communication between system elements during user interaction.



7.8 Communication Diagram

7.9 Package Diagram

The Package Diagram represents the logical organization of the SafeSpace system by grouping related elements into packages. It helps organize different website features according to their functionality. These features may include website navigation, support resources, safety planning, and legal information. The diagram provides a simplified view of the system's organization and makes its overall structure easier to understand.

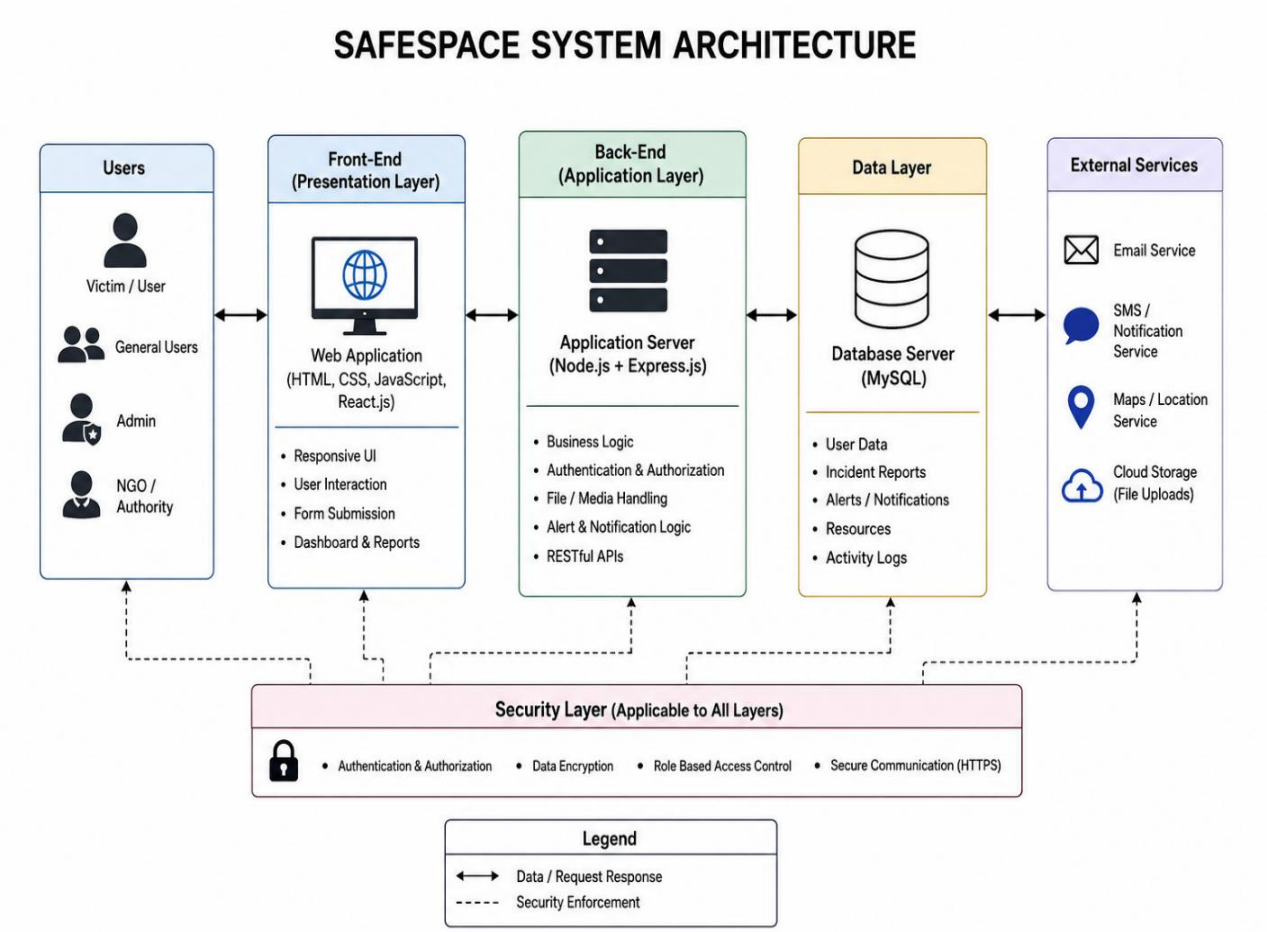


7.9 Package Diagram

Chapter 8. System Architecture Design

8.1 Overall Architecture

SafeSpace follows a **client-side/frontend-oriented architecture**.



8.2 Frontend

HTML

Responsible for:

- Page structure
- Navigation
- Content sections
- Buttons
- Forms
- Support modules

CSS

Responsible for:

- Layout
- Typography
- Colors
- Responsive design
- Disguised appearance

JavaScript

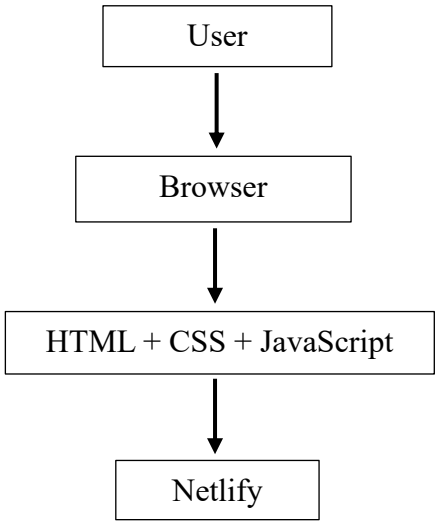
Responsible for:

- User interaction
- Dynamic functionality
- Hidden/triggered features
- Quick Exit
- Interactive support features

8.3 Backend

The current project documentation describes SafeSpace as a **frontend-only project**.
Therefore, a conventional application backend/database is not part of the current architecture.

Current Structure



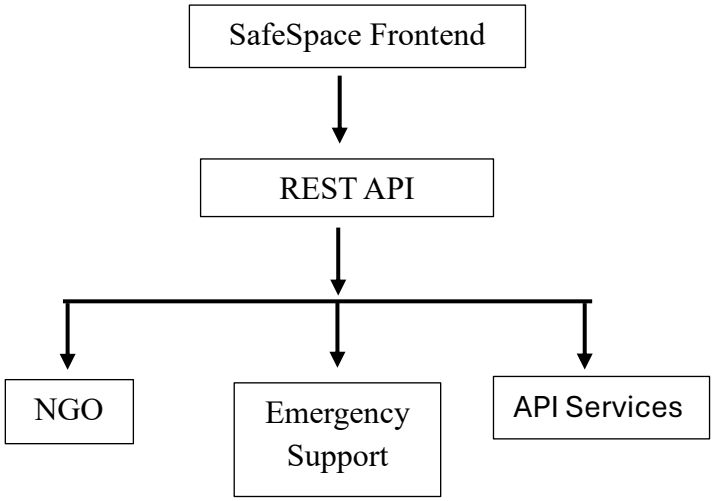
8.4 API Structure

No dedicated custom backend API is specified in the supplied project material.

Therefore, the current documentation should describe the API layer as:

API Layer: Not implemented in the current frontend-only version.Future API Structure

Future versions could potentially use APIs for:



The project documentation identifies future possibilities including NGO integration, emergency responders and AI-driven support.

Chapter 9. Implementation

9. Implementation

The SafeSpace website is implemented as a web-based frontend.

Implementation Modules

Module 1 – Neutral Homepage

The homepage is designed to look like a recipe website.

The live website currently displays **Fresh Recipes**, featured recipes such as Mediterranean Pasta Salad, Healthy Chicken Stir-Fry and Homemade Banana Bread, along with cooking tips. safespace-web.netlify.app

Module 2 – Support Interface

Provides access to the SafeSpace support functions.

Module 3 – Immediate Help

Provides emergency-support information.

Module 4 – Safety Planning

Provides a personalized safety-planning entry point.

Module 5 – Legal Information

Provides legal-rights and legal-option information.

Module 6 – Secure Contact

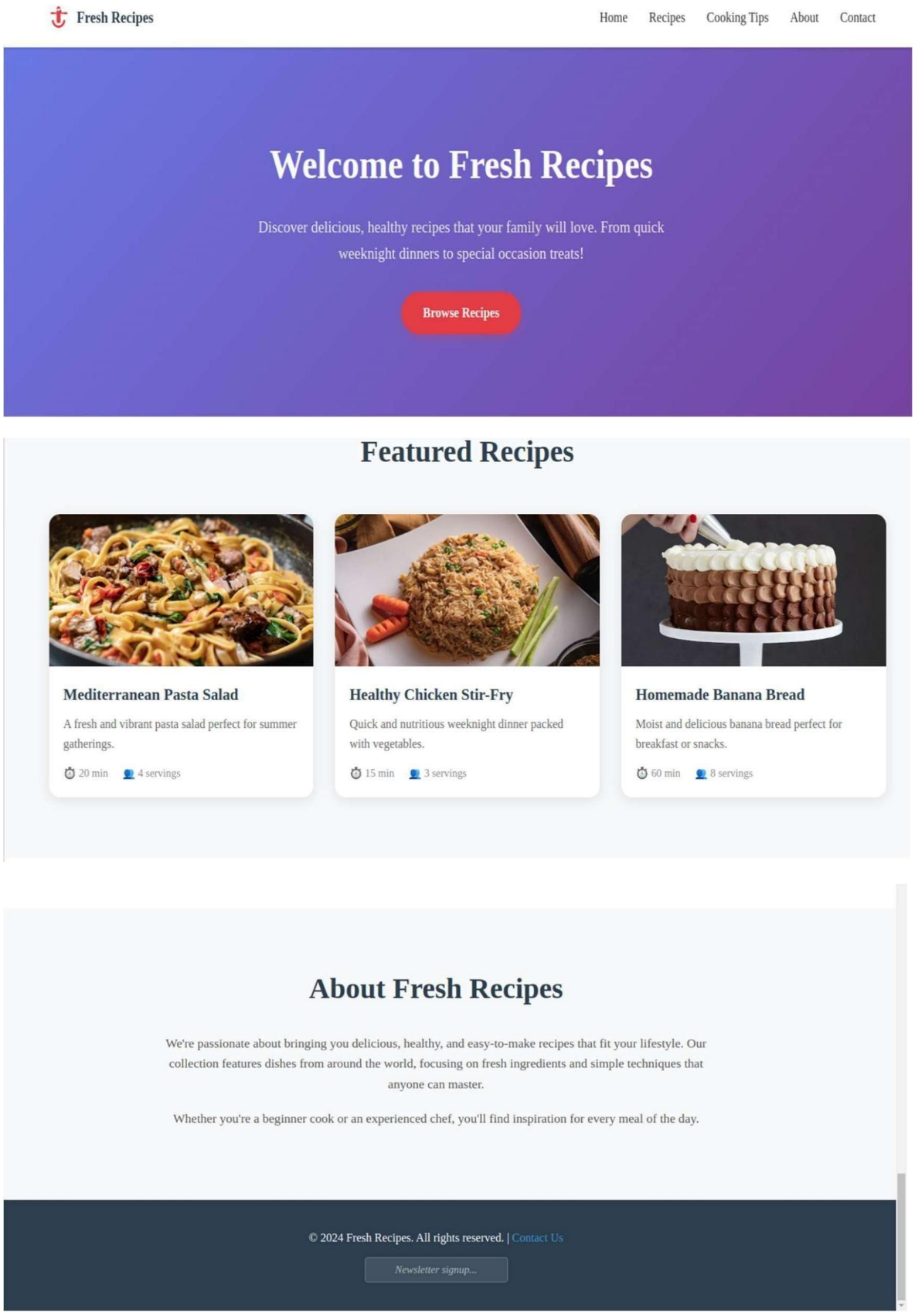
Provides access to support organizations.

Module 7 – Anonymous Chat

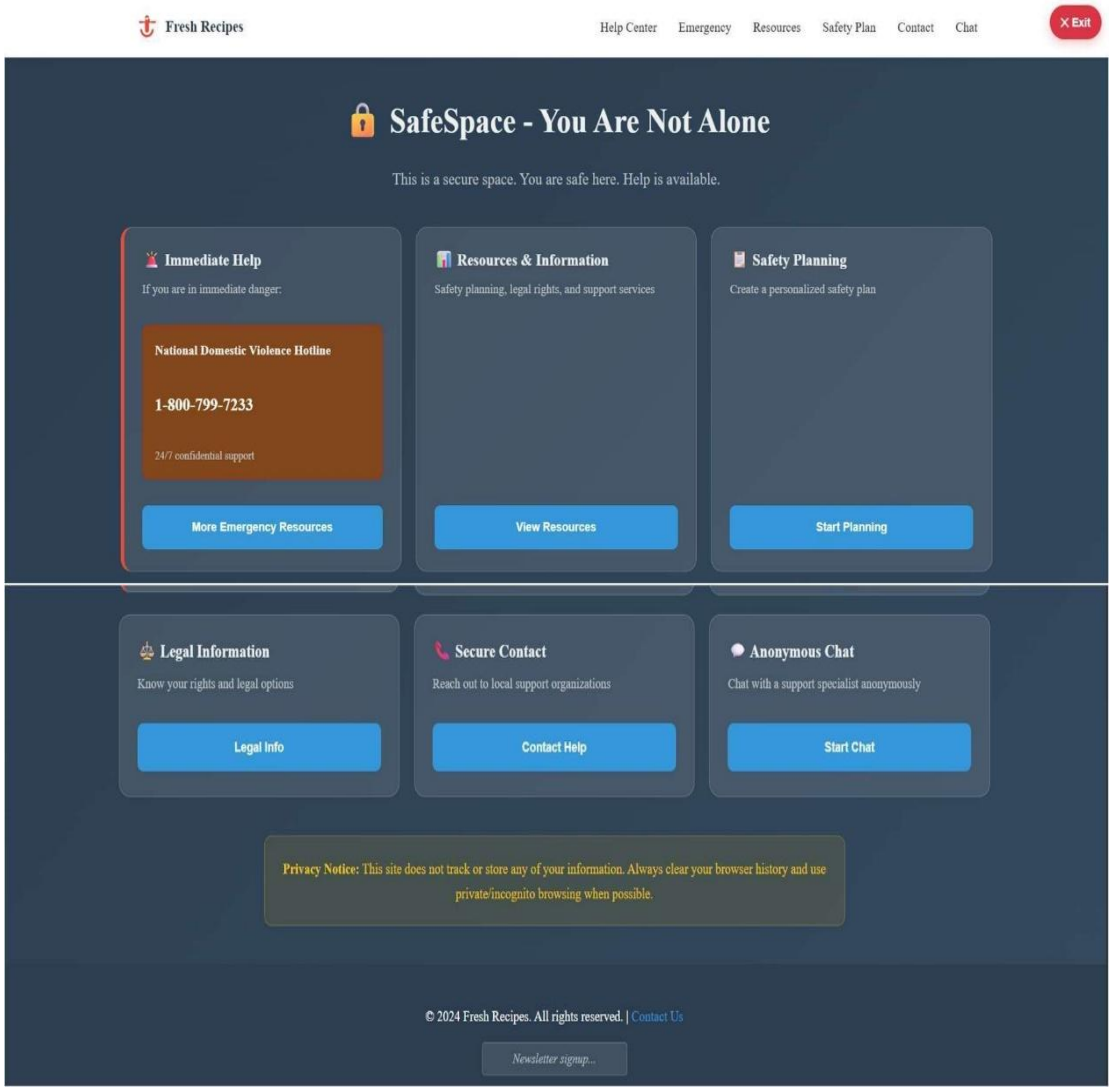
Provides an anonymous communication option. safespace-web.netlify.app

9.1 Screenshots of the Working System

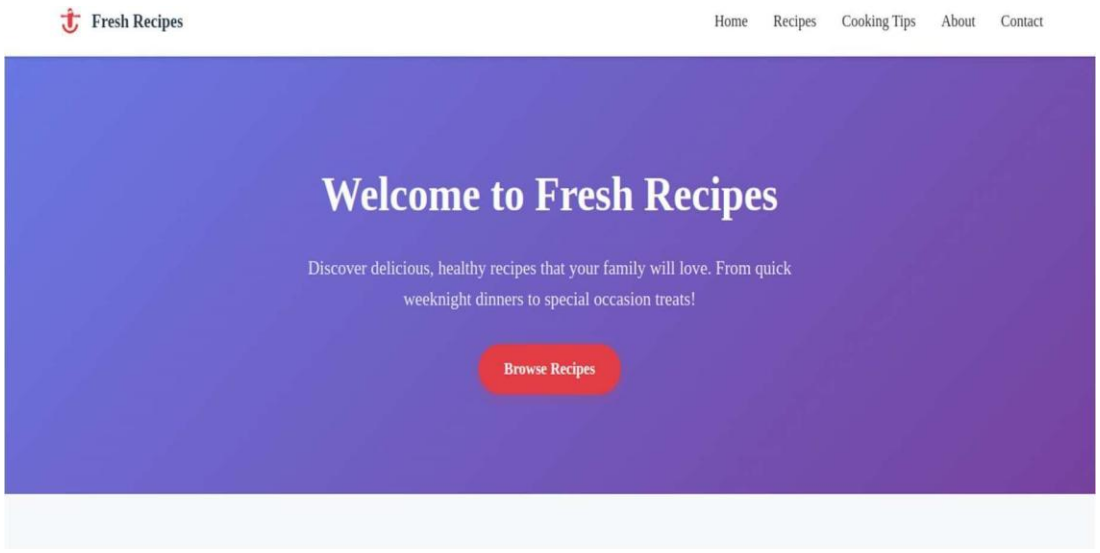
1. Neutral Homepage – Appears as a recipe



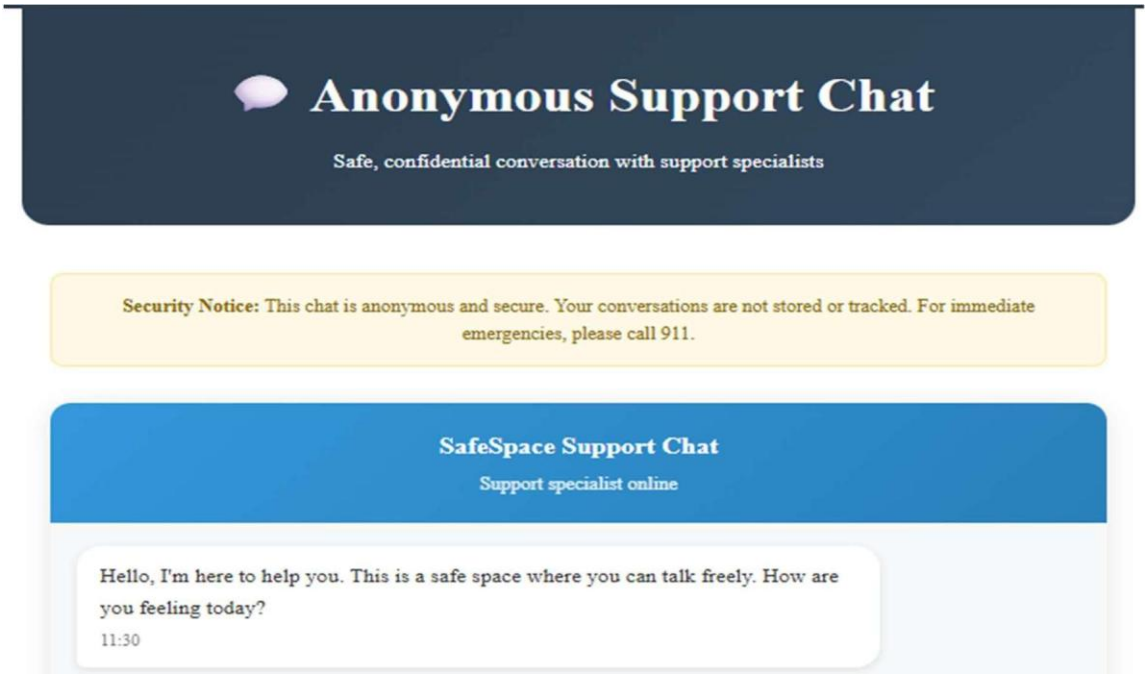
2. SafeSpace Interface – Displayed after entering secret code.



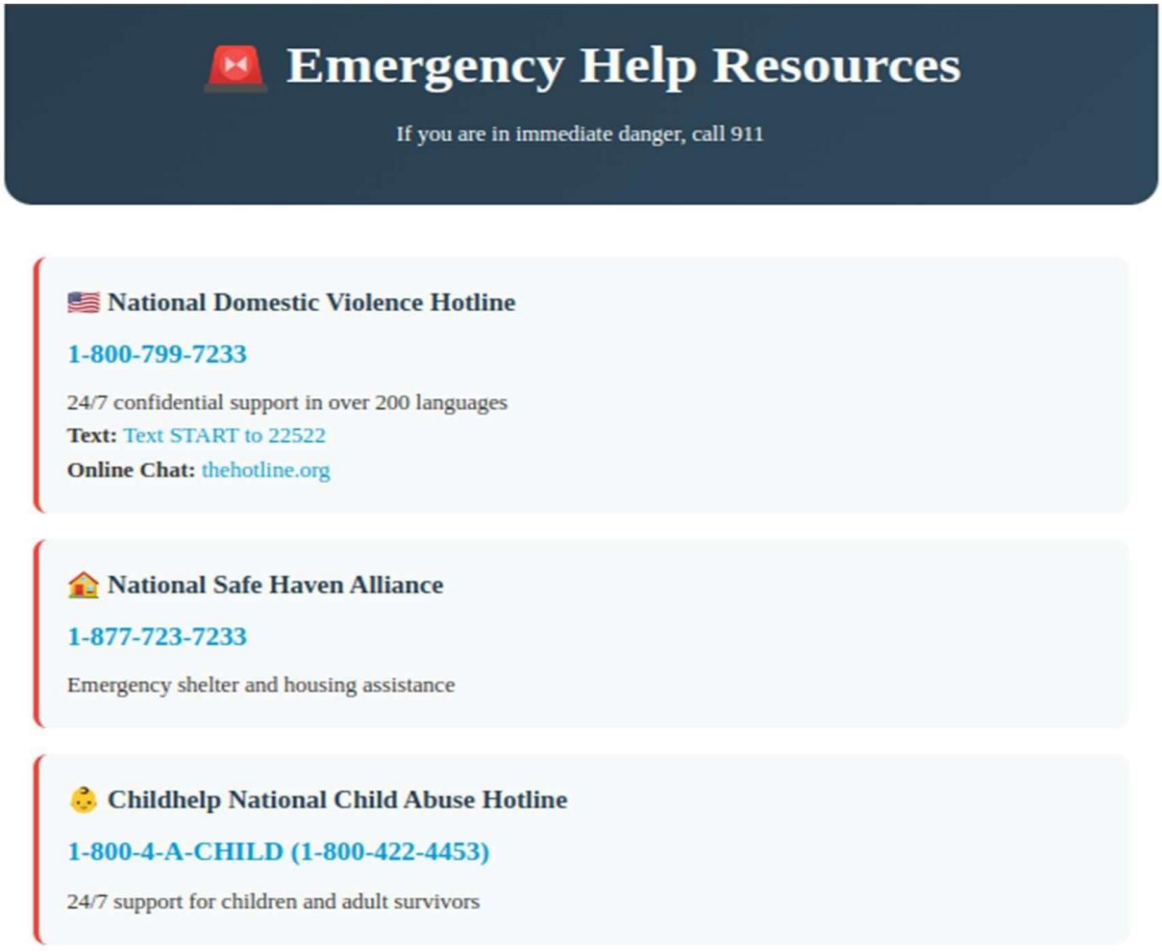
3. Quick Exit – Redirects to a neutral page.



4. Silent Chat – Shows note-like messages with auto-delete.



5. Code-Based Help – Example of coded emergency message.



🏳️‍🌈 **LGBT National Domestic Violence Hotline**

1-800-934-5280

Support for LGBTQ+ individuals experiencing abuse

🌍 **RAINN National Sexual Assault Hotline**

1-800-656-4673

24/7 support for survivors of sexual violence

If You Can't Call

Text for Help:
Text "LOVEIS" to 22522 for the National Domestic Violence Hotline

Online Chat:
Visit thehotline.org for live chat support

Email:
Many local organizations offer email support

Library Computers:
Use public computers if your devices are monitored

6. Resource Page – Helpline numbers, legal guidance, and exit strategy tips.

📚 Resources & Information

Knowledge and support for your journey to safety

Understanding Abuse

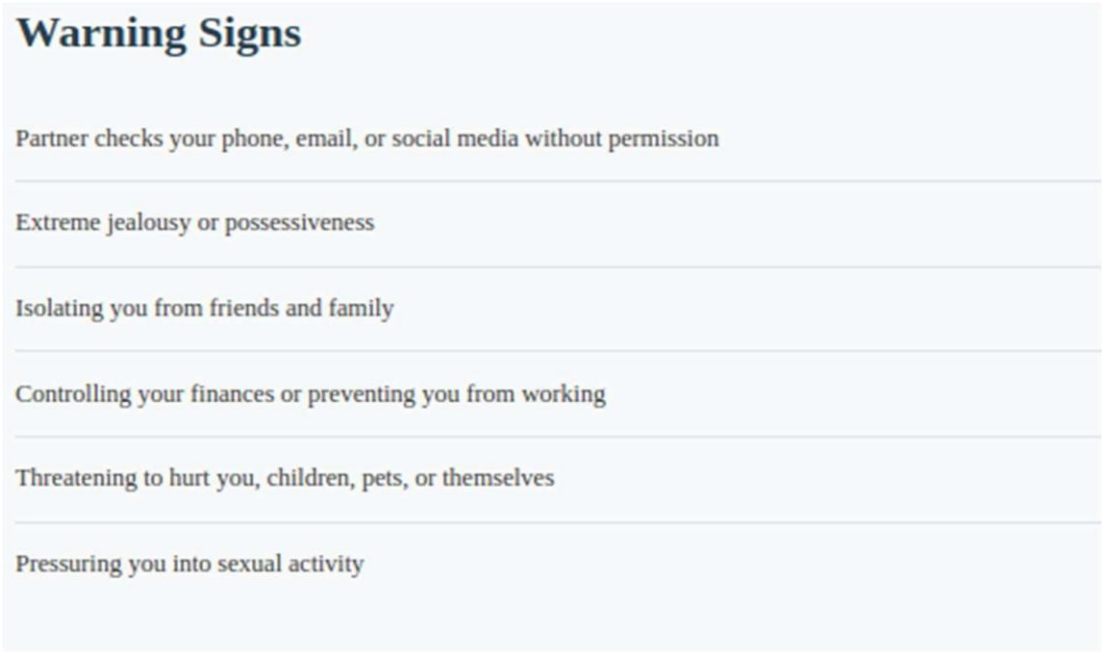
Physical Abuse:
Hitting, slapping, choking, restraining, or any unwanted physical contact

Emotional Abuse:
Name-calling, threats, intimidation, isolation from friends/family

Financial Abuse:
Controlling money, preventing work, hiding assets, ruining credit

Sexual Abuse:
Any unwanted sexual activity, including within marriage

Digital Abuse:
Monitoring devices, social media harassment, sharing private images





The implementation phase converts design into a working SafeSpace website. By using HTML, CSS, and JavaScript, the system provides a discreet, frontend-only platform that ensures privacy, anonymity, and safety. Key modules disguised entry, Quick Exit, silent chat, code-based help, and resources—work seamlessly to assist victims of domestic abuse in real-life scenarios. Screenshots and code snippets illustrate the practical functioning of the system. SafeSpace demonstrates how a well-designed frontend solution can be life-saving, secure, and accessible, even without a backend.

TESTING

9.2 Testing Strategy

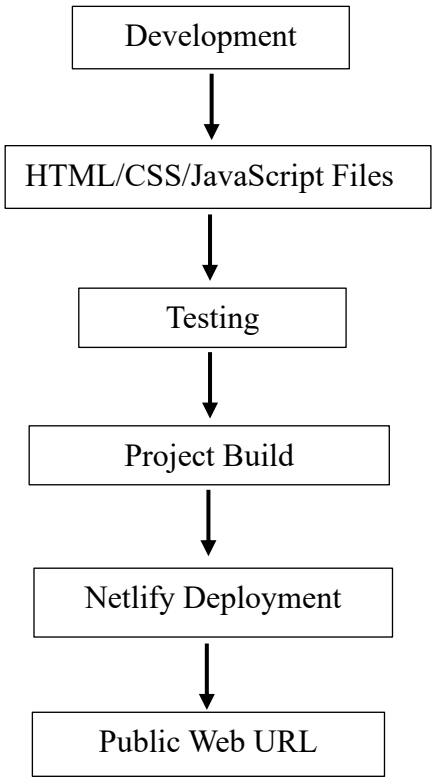
Test Case	Expected Result
Open Homepage	Recipe-style homepage
Click Home	Home page should appear
Click Recipes	Recipe section should open
Click Cooking Tips	Cooking tips should display
Access SafeSpace	Support section should become available
Immediate Help	Emergency information should appear
Resources	Support resources should appear
Safety Planning	Safety planning should open
Legal Information	Legal information should openb
Secure Contact	Contact functionality should open
Anonymous Chat	Chat interface should open
Quick Exit	User should be able to leave the support interface
Mobile Screen	Layout should remain usable
Desktop Screen	Layout should remain usable

Chapter 10. Deployment

10.Deployment

SafeSpace is deployed using **Netlify**. The project documentation specifies Netlify as the deployment platform.

Deployment Process



The current deployed website is:

SafeSpace Web: https://safespace-web.netlify.app/

The live page is accessible and currently presents the Fresh Recipes interface together with the SafeSpace support functionality. safespace-web.netlify.app

10.1 Evaluation

The project can be evaluated using the following parameters:

Functionality

The required support features should work correctly.

Usability

Users should be able to navigate the website without unnecessary complexity.

Responsiveness

The interface should remain usable on different screen sizes.

Privacy-Oriented Design

The neutral appearance is intended to reduce immediate disclosure of the website's support purpose.

Performance

The frontend-only approach keeps the architecture relatively lightweight.

Social Impact

The project aims to provide a discreet digital pathway to support and safety information.

The original project describes the overall objective as balancing **disguise, speed, safety and usability**.

Important Evaluation Note

The current live website itself displays a privacy notice saying that it does not track or store information and recommends clearing browser history/private browsing. safespace-web.netlify.app

For the final college documentation, these should be presented as **design/privacy claims of the project**, not as a guarantee of complete anonymity or absence of browser/network traces.

Chapter 11. Conclusion

11.Conclusion

SafeSpace is a web-based project designed to provide a discreet pathway to support for domestic-abuse victims.

The main concept is to combine a harmless-looking frontend with access to support-oriented functionality. The current website uses a recipe-style interface with features such as Fresh Recipes, Cooking Tips and About information, while the SafeSpace section provides emergency help, resources, safety planning, legal information, secure contact and anonymous chat. safespace-web.netlify.app

From a technical perspective, the project uses a lightweight frontend architecture based on HTML, CSS and JavaScript and is deployed through Netlify.

The project demonstrates how web technology and user-interface design can be combined to address a social problem while emphasizing discretion and ease of access.

11.1 Future Scope

SafeSpace has the potential to evolve into a complete life-saving ecosystem. Some areas of future development include:

1.Backend Integration for Real-Time Help

- Connect users directly with verified NGOs, legal aid centers, or police departments.
- Implement secure server-side chat with full end-to-end encryption.

2. AI-Powered Monitoring

- Use AI models to detect patterns of distress from messages.
- Suggest emergency actions automatically if high-risk situations are detected.

3. User-Defined Codes and Gestures

- Allow users to set their own secret codes, keywords, or gestures, making the system more personalized and secure.

4. Offline and SMS-Based Support

- Implement SMS-based emergency codes for areas with poor internet connectivity.
- Example: Sending a predefined SMS code to automatically alert NGOs.

5. Encrypted Vault for Sensitive Documents

- Provide an in-app encrypted gallery for storing evidence, ID proofs, or documents safely.

6. Mobile App Development

• Extend SafeSpace into a cross-platform mobile application with the same disguise and safety features.

7. Community Support System

• Add a survivor network forum (anonymous) where victims can support each other and share safe exit strategies.

Future versions may include:

1. NGO integration
2. Emergency responder integration
3. Secure backend services where appropriate
4. AI-assisted support
5. Geolocation-based SOS
6. Additional language support
7. Improved accessibility
8. More advanced safety-planning tools
9. Secure resource management
10. Integration with verified support organizations

Some of these future directions are already identified in the original project scope.

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